

Chapter 4, Problem 9RQ

Problem

A load is connected to a network. At the terminals to which the load is connected, $R_{Th} = 10\ \Omega$ and $V_{Th} = 40\text{ V}$. The maximum possible power supplied to the load is:

- (a) 160 W
- (b) 80 W
- (c) 40 W
- (d) 1 W

Step-by-step solution

Step 1 of 4

Draw the Thevenin equivalent circuit.

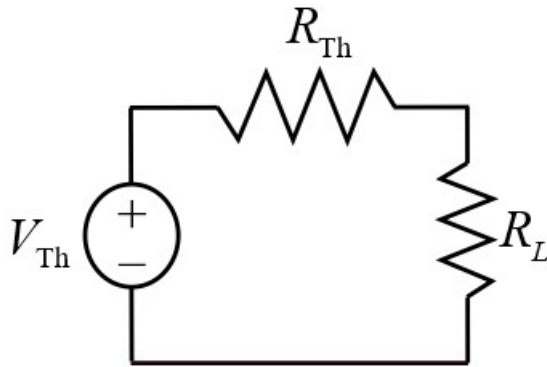


Figure 1

Step 2 of 4

Calculate the power supplied to the load.

$$p = i^2 R_L$$
$$= \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \dots\dots (1)$$

For the maximum power transfer, the load resistance should be equal to Thevenin's resistance.

$$R_{Th} = R_L$$

Step 3 of 4

Rewrite the equation (1).

$$\begin{aligned} p &= \left(\frac{V_{\text{Th}}}{R_{\text{Th}} + R_{\text{Th}}} \right)^2 R_{\text{Th}} \\ &= \left(\frac{V_{\text{Th}}}{2R_{\text{Th}}} \right)^2 R_{\text{Th}} \\ &= \frac{V_{\text{Th}}^2}{4R_{\text{Th}}} \end{aligned}$$

Step 4 of 4

Substitute 40 V for V_{Th} and 10Ω for R_{Th} .

$$\begin{aligned} p &= \frac{(40)^2}{4(10)} \\ &= 40 \text{ W} \end{aligned}$$

Hence, the option ☐ (c) is correct.